Activity 8— Advanced Numeric Pattern Generator

1. Problem Statement

Develop a Python program to generate a numeric pattern based on the number of rows entered by the user. Each row displays numbers from 1 up to the row number.

2. Objectives

- To generate numeric patterns using Python.
- To understand `for` loops and the `range()` function.
- To use nested loops.
- To improve basic programming logic.

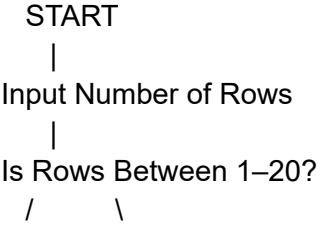
3. Assumptions

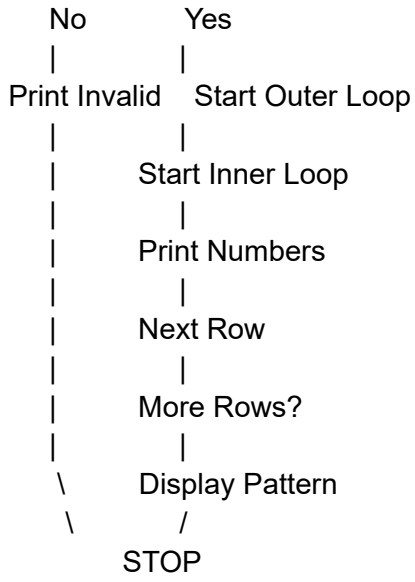
- The user enters the number of rows.
- The number of rows must be greater than zero.
- Each row displays numbers starting from 1 up to the current row number.
- For this example, the maximum number of rows is 20.

4. Algorithm

1. Start.
2. Read the number of rows.
3. Check whether the number is between 1 and 20.
4. Use an outer loop to control the rows.
5. Use an inner loop to print numbers in each row.
6. Move to the next line after each row.
7. Display the complete pattern.
8. Stop.

5. Flowchart





6. Pseudocode

```
START
INPUT rows

IF rows < 1 OR rows > 20
   PRINT "Invalid number of rows"
ELSE
   FOR i = 1 TO rows
      FOR j = 1 TO i
         PRINT j WITHOUT NEW LINE
      ENDFOR
      MOVE TO NEXT LINE
   ENDFOR
ENDIF

STOP
```

7. Python Source Code

File name: `activity8.py`

```
rows = int(input("Enter number of rows (1-20): "))

if rows < 1 or rows > 20:
   print("Invalid number of rows")

else:
   for i in range(1, rows + 1):
      for j in range(1, i + 1):
         print(j, end=" ")
```

print()

8. Test Cases

Input	Expected Output
3	1 / 1 2 / 1 2 3
5	Pattern with 5 rows
1	1
0	Invalid number of rows
21	Invalid number of rows

9. Output Screenshot and Conclusion

Sample Output:

Enter number of rows (1-20): 5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

Output Screenshot: Run `activity8.py` in Python IDLE or VS Code. Capture the input and output, and save the screenshot as `activity8_output.png`.

Conclusion: This program generates a numeric pattern using nested `for` loops and helps develop basic Python programming logic.

INPUT:

```
rows = int(input("Enter number of rows (1-20): "))

if rows < 1 or rows > 20:
    print("Invalid number of rows")

else:
    for i in range(1, rows + 1):
        for j in range(1, i + 1):
            print(j, end=" ")
        print()
```

OUTPUT:

```
Enter number of rows (1-20): 3
1
1 2
1 2 3

========== RESTART: C
Enter number of rows (1-20): 5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

========== RESTART: C
Enter number of rows (1-20): 1
1

========== RESTART: C
Enter number of rows (1-20): 0
Invalid number of rows

========== RESTART: C
Enter number of rows (1-20): 21
Invalid number of rows
```